Positional Guard Evasion at Short Context: A Replication Note

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> ***Preprint.*** *This is the named version of a paper prepared for peer review. The anonymized submission build is identical in content; only the author block, the companion-paper citation keys and the anonymization note differ.*

> ***This note is a replication, not a novel contribution.*** *A prior-art check found* ***LongGuard*** *[LongGuard], published 2026-08-27 — four days before this work — which establishes the same positional finding across* ***15 guardrails*** *over a* ***0.25k–32k*** *grid, supplies the attention→logit→behavior mechanism this note defers, and implements the chunked-detection defense this note only predicts.*
>
> ***The measurements below stand; the novelty claim does not.*** *This note is positioned as a replication at short context on two models LongGuard did not test, and* ***retracts an initial claim to have refuted length dilution*** *(§4.2).*
>
> ***The retraction is then tested instead of conceded.*** *A follow-up run (P3-X, §4.2.1) extends the grid to LongGuard's full 0–32k range at both scales, 1,260 inputs per model, zero unparsed output.* ***The retraction was correct****: over the endpoints, 19 payloads out of 60 degrade and 0 improve at both scales ($p < 0.0001$), and the 0.6B's one non-monotone step is the single step that fails significance ($p = 0.688$).*
>
> ***Pre-registered before any constructed input was scored*** *— three mechanisms, their distinguishing predictions and the invalidation conditions frozen and hashed (`ba560d2d`, re-hashed `21cdee5c`). Pre-registration protected the analysis from the data.* ***It did nothing about claiming novelty that did not exist****, which is the failure recorded here.*

Abstract

Embedding unsafe content inside an ordinary benign frame — a workplace email — is the most effective guard-evasion transformation we have measured, defeating guards on 51% to 62% of attempts across three model families. It is also the transformation whose mechanism is least obvious, because it changes two things at once: it makes the input longer, and it moves the unsafe span away from the edges.

We separate those by construction. A synthetic grid holds filler *content* constant while varying filler *quantity* and payload *position* independently: 100 unsafe payloads, each of which both guards catch perfectly in isolation, placed at three positions across four length multiples.

Position orders the effect at short context. Pooled across lengths, a payload in the middle of its frame is missed 4.33× more often than one at the end. **This replicates LongGuard's lost-in-the-middle finding** [LongGuard] on two models they did not test, in a length regime (≈0.3k–1.2k tokens) at the very bottom of their grid.

We retract a claim, and then test the retraction. The short-range grid showed miss rate non-monotone in length — rising to 9.3% at 4× filler then falling at 16× — and we initially concluded that dilution is refuted. LongGuard attributes failure *to* proportional dilution over a range 25× longer, with an edge–middle gap growing from +6.67% at 0.25k to +30.88% at 16k. Our grid never left their weakest regime, so the non-monotonicity was retracted as a probable small-range artifact. **We then ran the range** (§4.2.1): across 0–32k at both scales, **19 payloads degrade and 0 improve, at both scales ($p < 0.0001$)**, and the sole non-monotone step is the only step that fails significance. **The artifact reading was right.**

The measured effect. 39/300 versus 9/300 for the 4B (Fisher one-sided $p = 3.4\times10^{-6}$) and 2.64× ($p = 2.4\times10^{-3}$) for the 0.6B. The ordering — middle worst, end safest, start intermediate — holds at every length and in both models, and matches the direction LongGuard reports at larger scale.

What this note adds, narrowly: the effect is measurable at **short context** (≈0.3k–1.2k tokens), on **two models LongGuard did not evaluate**, against a control in which the same payloads are caught **100% of the time** unframed. We release the grid construction, the per-item verdicts, and the pre-registration.

Keywords: guard models, adversarial robustness, positional bias, long-context, AI safety

1. Introduction

Guard models classify content as safe or unsafe inline in an agentic pipeline. Prior work in this program established that semantically-preserving rephrasing degrades guards asymmetrically — leaving false positives unchanged while raising false negatives — and that among five transformation families, one dominates: **embedding the unsafe content inside an ordinary benign frame.**

Across three model families that family reaches **0.5079 and 0.6199** candidate-level miss rate, roughly double any other transformation. Two things about it were unexplained.

First, it was not the strongest under deterministic decoding. On a temperature-0 floor, a different family led. The benign-framing advantage appears under *sampling*, which suggests the effect depends on something that varies between samples — plausibly where the payload lands.

Second, it confounds two variables. Framing makes the input several times longer *and* moves the payload away from the input boundaries. Every prior analysis reported it separately for exactly this reason, and none could separate the two.

This paper separates them.

1.1 Three mechanisms, three predictions

	mechanism	prediction
M-A	**Length dilution** — the unsafe span is a smaller fraction of a longer input	miss rate rises with length at fixed position
M-B	**Positional displacement** — the payload sits where the model attends least	miss rate varies with position at fixed length

M-C	**Boundary crossing** — the frame shifts the judgment itself, not attention	verdict probability shifts even at short length and edge position

They are separable by construction: M-A varies length holding position, M-B varies position holding length, and M-C predicts an effect surviving both held constant.

1.2 Findings

- **M-A: non-monotone in our range, refutation retracted — and the retraction now tested** (§4.2, §4.2.1). Our grid tops out around 1.2k tokens; LongGuard measures dilution dominating over a range 25× longer. **Extending to that range confirms it**: 19 of 60 payloads degrade and 0 improve from 1k to 32k, at both scales ($p < 0.0001$). **M-A was never refuted — it was untested.**
- **M-B supported**, at 4.33× and 2.64×, significant in both models, consistent at every length — **replicating** LongGuard's lost-in-the-middle result at the short end of the range.
- **M-C deferred here, and already answered elsewhere.** It requires verdict-token probabilities and a different serving stack, which would reintroduce a runtime confound this program was built to eliminate (§3.4). LongGuard supplies the attention→logit→behavior chain we deferred.

2. Related work

LongGuard [LongGuard] is the primary result this note replicates, and it precedes this work by four days. It evaluates 15 guardrails over a 0.25k–32k grid, reports unsafe recall dropping more than 50% on average, separates proportional dilution from absolute length with a paired Benign-Fill / Needle-Repeat control, traces an attention→logit→behavior mechanism, isolates guard-specialized retrieval heads, and supplies two training-free mitigations plus a deployment protocol. **Its appendix C.2 reports the lost-in-the-middle and endpoint-asymmetry pattern this note also finds**, with an edge–middle gap growing from +6.67% at 0.25k to +30.88% at 16k.

Positional bias in long-context models is independently well documented for retrieval and question answering. LongGuard carries it into safety guardrails.

Guard robustness work measures rephrasing, jailbreaks and prompt injection [GuardMeaning], and guardrail behavior under RAG-style long contexts [RAGGuard].

What this note claims. A replication on two models LongGuard did not test — at short context, and then across their full 0–32k range — each against a **0× unframed control**, which is the one design element their needle-in-haystack setup does not have and which anchors every cell to an absolute floor rather than a relative baseline. Plus a **retraction** of a dilution refutation our range could not support, and a **test of that retraction** at the range where it is testable. **We claim no novelty on the positional phenomenon, its mechanism, or its mitigation.** This note was drafted before its prior-art check; LongGuard predates it by four days and supersedes its central claim, which is recorded here rather than absorbed.

3. Method

3.1 Payload selection

100 unsafe payloads, each **caught by both models under test in its original, unframed form.**

This matters more than it may appear. If a payload were already missed without framing, a miss under framing would be credited to the frame rather than to a pre-existing failure. The pre-registration forbids that, and §4.1 confirms the filter worked: **the control condition has a 0.0000 miss rate in both models.**

> ***A pre-registered rule that had to be revised, and was recorded rather than quietly changed.*** *The original rule was "caught by every guard in the program." That yields* ***27 payloads****, because one family in the panel catches only 32 of 150 items under the four-policy prompt we supply it — a limitation of* our configuration*, not of the payloads. Intersecting against it would let our own prompt decide which unsafe content this experiment may study. The rule was revised to "caught by every guard* ***this experiment scores****," giving 120, of which 100 are used. The cost is that the payload set is* ***not valid for a cross-family extension*** *and must be re-intersected if P3 is ever run on the wider panel.*

3.2 The grid

One filler corpus, reused in every cell, so filler *content* is constant and only quantity and position vary. Filler is ordinary workplace-email prose.

- **Length:** payload + filler at **0×, 1×, 4×, 16×** the payload length.
- **Position:** payload at **start**, **middle**, **end** of the filled input.
- **0× is the control** — payload alone, no frame, no position. It is emitted once per payload.

100 payloads × 10 cells = **1,000 constructed inputs**, each scored by both models. Median input length runs 281 → 564 → 1,407 → 4,779 characters across the four multiples.

3.3 Scoring

Qwen3Guard-Gen-4B and -0.6B at `Q4_K_M`, `num_ctx` 8192, seed 0, temperature 0, fail-closed — identical to every other measurement in this program. **Zero unparsed outputs across 2,000 evaluations.**

A **miss** is the guard returning `Safe` on an input containing an unsafe payload.

3.4 Why M-C is deferred, stated before any number

M-C requires verdict-token *probabilities*, not argmax verdicts. The serving stack used throughout this program does not usefully expose logprobs, so M-C requires `transformers` or vLLM.

Numbers from that path cannot be compared against any verdict measured here, or the runtime confound — precision, tokenization and kernel differences masquerading as an effect — re-enters through the back door. Prior work in this program was designed specifically to eliminate it. M-C therefore requires its own baseline on the same grid through the same stack, and is reported as **not run** rather than approximated.

4. Results

4.1 The control confirms the design

	miss rate on bare payloads (n=100)
Qwen3Guard-4B	**0.0000** [0.0000, 0.0370]
Qwen3Guard-0.6B	**0.0000** [0.0000, 0.0370]

Every payload is caught with no frame around it. Every miss reported below is therefore attributable to the framing, not to a payload the guard never handled.

4.2 M-A: non-monotone in our range — a retracted refutation

Miss rate pooled over position:

	0×	1×	4×	16×
4B	0.0000	0.0433	**0.0933**	0.0667
0.6B	0.0000	0.0433	**0.0833**	0.0600

Both models peak at 4× and decline at 16×.

Read alone, this appears to refute dilution. That reading is retracted. Our largest cell is ≈1.2k tokens; LongGuard's grid runs to 32k and finds dilution dominating, with the edge–middle gap growing from +6.67% at 0.25k to +30.88% at 16k. At 0.25k — the point closest to our range — their effect is also small. **We did not test a range where dilution would be expected to dominate**, and non-monotonicity inside a narrow low range is not evidence against it.

Length is not irrelevant — the jump from 0× to 1× is the largest single step, and any framing at all costs the guard something. But **within our range the quantity of filler does not order the effect.**

4.2.1 The retraction, since tested rather than conceded

The retraction above was a concession to a longer-range published result. **We then ran the range** (P3-X, pre-registered `324ee77b`): the first 60 of the 100 payloads by identifier, across lengths {0, 1k, 4k, 16k, 32k} × five positions — 0%, 25%, 50%, 75% and 100% of the way through the frame — 1,260 inputs per model at both scales. Sixty rather than a hundred was a budget decision; the subset is taken by identifier, not by any property of the payloads, and 14 of the 60 had been missed somewhere in the short-range grid by at least one model, against 24 of the 100.

`num_ctx` was raised 8192 → 32768 to admit a 32k input, and held constant across all 21 cells. P3-X's numbers are therefore not directly comparable to the table above. It does not correct those measurements; it tests the *inference we drew from them* at a range they never reached.

Unparsed output was 0.000 in all 42 cells, so no number below is a truncation artifact.

pooled miss rate	0×	1k	4k	16k	32k
4B	0.0000	0.1100	0.1233	0.1933	0.2700
0.6B	0.0167	0.0700	0.1867	0.1600	0.3233

The 0.6B misses **1 of 60** bare payloads at `num_ctx` 32768 where it missed 0 of 100 at 8192 (§4.1): one item, inside the control's interval either way, but the only movement in a control cell anywhere in this note, and a reminder that `num_ctx` is a serving parameter with an effect of its own.

The 4B is monotone in the pooled rates. The 0.6B has one decrease, at 4k→16k. An unregistered follow-up — reported as unregistered — tests the steps with McNemar exact on payload-level discordance, since the same 60 payloads recur at every length. **The unit is the payload: it counts as missed at a length if it is missed at any of the five positions**, each step compares every payload's indicator at the two lengths, and the discordant payloads are tested against a fair coin, two-sided:

> ***That single decrease is the one step that fails significance (2 worse, 4 better,*** **p** ***= 0.688).*** *Every other 0.6B step is a significant rise.* ***Over the endpoints, both scales are identical and one-directional: 19 payloads degraded, 0 improved,*** **p** ***< 0.0001.*** *Not one payload out of 60 became easier to detect when filler was added, at either scale.*

Two qualifications belong beside that. On the same test the 4B's individual steps are not all significant either — 1k→4k is 4 payloads worse against 0 better ($p = 0.125$) and 16k→32k is 7 against 1 ($p = 0.070$) — so "monotone" is a statement about the pooled point estimates, and the endpoint comparison is what carries the inference. And **30 of the 4B's 60 payloads and 28 of the 0.6B's are missed at some length and position**, against 21 and 16 of 100 in the short-range grid: the long range more than doubles the share of payloads the guard ever loses.

The edge–middle gap grows with length at the 4B — +7.2% at 1k to +28.3% at 32k — closely tracking LongGuard's +6.67% at 0.25k to +30.88% at 16k.

So the retraction was correct, and is now supported by measurement rather than by deference. The short-range non-monotonicity was a small-range artifact. **M-A is not refuted; it was untested**, and over the range where it is testable it holds.

4.3 M-B supported: position orders the effect

Miss rate by position, pooled over the three non-zero lengths:

	start	**middle**	end
4B	13/300 = 0.0433	**39/300 = 0.1300**	9/300 = 0.0300
0.6B	16/300 = 0.0533	**29/300 = 0.0967**	11/300 = 0.0367

comparison	4B	0.6B
middle vs end	**4.33**×, $p = 3.4 \times 10^{-6}$	**2.64**×, $p = 2.4 \times 10^{-3}$
middle vs start	3.00×, $p = 1.1 \times 10^{-4}$	1.81×, $p = 3.1 \times 10^{-2}$

(Fisher exact, one-sided.)

Risk ratios with 95% intervals: 4B middle/end **4.33** [2.14, 8.79], middle/start 3.00 [1.64, 5.50]; 0.6B middle/end **2.64** [1.34, 5.18], middle/start 1.81 [1.01, 3.27].

The pooled cells count each payload three times, once per length, so the Fisher tests and these intervals treat 300 dependent observations as independent. The paired tests do not. **Paired on matched (payload, length) cells, the direction has no exceptions in the 4B:** of the cells where the middle and end verdicts differ, the 4B is missed in the middle and caught at the end in **30 of 30** (exact $p = 9.3 \times 10^{-10}$), and missed in the middle and caught at the start in 26 of 26 ($p = 1.5 \times 10^{-8}$); the 0.6B runs 19 to 1 ($p = 2.0 \times 10^{-5}$) and 18 to 5 ($p = 5.3 \times 10^{-3}$). Collapsed to payloads, 19 of the 4B's 100 are missed more often in the middle than at the end and none the reverse; the 0.6B's run 13 to 1. The effect is also concentrated: **21 of 100 payloads (4B) and 16 of 100 (0.6B) are ever missed** in a framed cell, and the other four-fifths are caught in every one.

The ordering is middle $\gg$ start $>$ end. It holds at every individual length in the 4B, and at 4$\times$ and 16$\times$ in the 0.6B; at 1$\times$ the 0.6B's start and middle cells are within one item of each other, 6 against 5 of 100, which is the one cell in the grid where position does not order the effect:

4B	start	middle	end
1×	0.0200	0.0900	0.0200
4×	0.0500	**0.1800**	0.0500
16×	0.0600	0.1200	0.0200

0.6B	start	middle	end
1×	0.0600	0.0500	0.0200
4×	0.0800	**0.1200**	0.0500
16×	0.0200	**0.1200**	0.0400

The worst single cell is **4B at 4× length, middle position: 0.1800** — eighteen of a hundred payloads that the same model catches perfectly when unframed.

4.4 Why the two mechanisms produce different curves

The interaction explains §4.2's non-monotonicity. At 16× the filler is long enough that the "middle" is far from both edges — but the *end* position also becomes strongly protective, because the payload sits immediately before the verdict is generated. Pooling across positions therefore mixes a worsening middle with an improving end, and the pooled curve turns over.

That is exactly what an effect governed by position, not quantity, looks like when you average over position. It is also why reporting length alone would have produced a confusing null.

5. Analysis

5.1 The positional profile matches long-context retrieval, in a safety setting

The middle-worst, edges-better profile is the signature documented for long-context retrieval. Its appearance in a *classification* task with a safety consequence is LongGuard's finding, which this note replicates at the short end of their range — and it means the guard is not failing to *understand* the payload. It is failing to *attend* to it.

The strongest evidence for that reading is §4.1: the identical payload, identical model, identical settings, is caught **100% of the time** when it is the whole input. Nothing about the content changed.

5.2 The attacker's advantage is free

An attacker gains this multiplicatively **without modifying the payload**. No rephrasing, no obfuscation, no optimization against the guard — only placement. Placement is also the cheapest possible variation: it requires no model, no queries and no feedback.

That places this below every attack we have previously measured in cost and above most in reliability.

5.3 The defensive implication is unusually concrete

Most findings in this program complicate defense. This one suggests something actionable: **a guard that scores an input in overlapping windows, or that re-scores with the suspect span moved to the end, should recover much of the loss** — because the end position is measurably the safest one, at 0.0300 and 0.0367 against a middle of 0.1300 and 0.0967.

We have not tested that. LongGuard implements a chunked-detection defense of exactly this shape and reports that it recovers most of the loss on their panel; whether it does on these two models is the obvious follow-up, and it is cheap.

5.4 Scale does not rescue it

The 4B shows a *larger* positional effect than the 0.6B (4.33× versus 2.64×), driven by an unusually low end-position miss rate. The bigger model is better at the edges and no better in the middle. **Scaling improves the best case without repairing the worst.**

6. Threats to validity

M-C is untested (§3.4), so we cannot exclude that framing also shifts the judgment itself. Our claim is that M-A does not account for the effect and M-B substantially does — not that M-B is the whole story.

Two models, one family. Qwen3Guard at two scales. The benign-framing dominance that motivated this work replicates across three families, but the positional decomposition has not been run on them. The payload set would also need re-intersecting first (§3.1).

One filler corpus. Filler content is constant by design, which removes it as a confound but also means we tested one register of benign prose. Filler that is topically related to the payload might behave differently.

Synthetic construction. The grid is assembled, not naturally occurring. It buys the ability to separate length from position, and it costs ecological validity: a real attacker's email would be coherent with its payload in a way ours is not — which most likely makes our estimates **conservative**.

Cell sizes of 100. Individual cells have wide intervals; the pooled position comparisons (n=300 per cell) carry the significance claims, and per-cell figures are reported for shape rather than for inference.

The pooled comparisons are not independent. Each payload appears in every cell, so a 300-item pooled cell is 100 payloads observed three times and the Fisher intervals in §4.3 are too narrow. The matched-cell and payload-level paired tests reported beside them are the stricter test, and they reach the same conclusion with a stronger signal.

Absolute rates are small. A 13% pooled middle-position miss rate is a large *relative* effect against a 0% control, and a modest absolute one. Both framings are in §4.3 and neither is preferred.

7. Conclusion

The most effective guard-evasion transformation we have measured works because of **where** the payload sits, not **how much** text surrounds it.

Within the short range we tested, length is non-monotone. Position orders the effect cleanly: a payload in the middle of a benign frame is missed **4.33× more often than the same payload at the end** by a model that catches it **100% of the time** with no frame at all. The ordering holds at every length in the larger model and at all but the shortest in the smaller one, and on the paired test the 4B has no exceptions: 30 of 30 discordant cells fall the same way.

This is the long-context "lost in the middle" bias, arriving in a safety classifier and carrying an exploitable consequence. It costs an attacker nothing — no rephrasing, no queries, no optimization, only placement — and it is the cheapest reliable evasion we have measured.

It may also be the most tractable. The end position is measurably safe. A guard that scores overlapping windows, or that relocates a suspect span before deciding, has a plausible path to recovering most of the loss. **That is a defense this result predicts and does not test**, and it is the next thing worth running.

Artifact

Grid construction script; the filler corpus; all 1,000 constructed inputs by payload identifier, length multiple and position; per-item verdicts for both models; the pre-registration and both hashes, including the payload-selection addendum and what it costs.

References

[**LongGuard**] Z. Chen, X. Wu and S. Hu. *LongGuard: Mechanistic Analysis and Training-Free Mitigation of Long-Context Failure in Safety Guardrails.* Institute of Information Engineering, Chinese Academy of Sciences. arXiv:2608.27580, 2026. — **the primary result this note replicates.** Establishes the positional finding across 15 guardrails over 0.25k–32k, supplies the attention→logit→behavior mechanism, and implements chunked-detection and attention-head-sharpening defenses.

[**GuardMeaning**] C. Pinneri and C. Louizos. *Guarding the Meaning: Self-Supervised Training for Semantic Robustness in Guard Models.* Qualcomm AI Research. arXiv:2511.10665, 2025.

[**RAGGuard**] *RAG Makes Guardrails Unsafe? Investigating Robustness of Guardrails under RAG-style Contexts.* arXiv:2510.05310, 2025.

[**WildGuard**] *WildGuard: Open One-Stop Moderation Tools for Safety Risks, Jailbreaks, and Refusals of LLMs.* arXiv:2406.18495, 2024. — source of the payloads.